HOME EXAMINATION

FIE401

Spring, 2026

Start: *11 May, 09:00*

End: *13 May, 14:00*

THE HOME EXAMINATION SHOULD BE SUBMITTED IN WISEFLOW

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Ouriginal, a plagiarism control system used by NHH. Use of artificial intelligence (AI) in this exam is determined by the course coordinator. Information on AI usage and guidelines can be found on the course's Canvas page.

SUPPLEMENTARY REGULATIONS FOR HOME EXAMINATIONS

You can find supplementary regulations under the headline "Regulations"
https://www.nhh.no/en/for-students/regulations/
Find more information under chapter 4.0 in the Supplementary provisions to the regulations for fulltime study programmes

Number of pages, including front page: *4*

Number of attachments: *3*

FIE401 - Final Project

Spring 2026

Formalities

This final project will be handed out on **11.05.2026 at 9:00** and has to be submitted no later than the **13.05.2026 at 14:00** via WiseFlow.

In addition to the required hand-in files, there will be an oral exam on the **02.06.2026 – 04.06.2026.** Please sign up for time slots on CANVAS as a group. You do not need to be present at the presentations of your peers other than your own group. You will have strictly 5 minutes to present your solution which will be followed by a question and answer session lasting up to 20 minutes. The latter will refer to **both your analysis of the final project as well as any other topics covered in class.**

This exam is group-based (three-four people per group). All group members will receive the same grade for the written submission, while grades on the oral exam will be individual.

Groups must specify in the first lines of the code whether they developed the code with/without using chatGPT or other generative AI. Groups must specify whether and how they used chatGPT or other generative AI while writing a report.

Please submit the following documents

- **Coding file (.R)** Please comment your general coding approach for each task. You do not need to explain the used functions.
- **Report (.pdf)** In the report, you should present, analyse, and interpret your results. You should provide numeric and written answers to all questions asked in the exam. Do not discuss your coding in the pdf as this should be found in the R file. Please keep your answers short and precise and focus on the questions asked in the outline of the project.
- **Presentation (.pdf)** It is not allowed to change the slides between your submission and the actual presentation. Make sure that you use your 5 minutes presentation time wisely. It is important that you provide an economic rational for choices you made during this project.

Formal requirements

- **Report**
  - Motivate all your choices.
  - Interpret your findings from a statistical and economic perspective.
  - Your report should read like a mini thesis/article having following structure:
    1. Abstract - summarise your findings (word limit = 100 words).
    2. Intro - outline the research objective and findings (max. three paragraphs, approx. one page).
    3. Data - summarize the data and variables used (Table 1).
    4. Analysis - your analysis (Table 2 & 3, and potentially Figure 1).
    5. Conclusion - your findings destilled.
  - The report should include three tables and one (optional) figure.
  - The maximum word count of the report without title, tables, abstract, and header is 3000.
  - Make sure that tables are self-contained and as described in the last lecture.
  - Present a maximum of six models in Table 2, and six models in Table 3.

- Use fairly plain formatting ensuring readability.
  - No appendix.
  - You do not need to explain the empirical models used as this will be part of the oral exam.
- **Presentation**
  - The main purpose of the presentation is to be a starting point for the oral exam.
  - The time limit of 5 minutes will be strictly enforced.
  - Discuss your choices of how you set up the econometric analysis.
  - All three tables of the report have to be in the presentation.
  - You do not need to prepare an appendix for the presentation.
  - Do not write your names or any identifying information on the slides.

Your task

In this final project, you will investigate the following research question:

How does market fragmentation affect stock's liquidity?

Trading of the same security across multiple venues (commonly referred to as market fragmentation) is a defining feature of modern equity markets. Its impact on market liquidity, however, is ambiguous. On the one hand, fragmentation fosters competition among trading venues, which can reduce execution costs. It also tends to attract high-frequency traders, who often act as de facto liquidity providers and may contribute to tighter bid–ask spreads. On the other hand, fragmentation disperses trading activity across venues, which can make it more difficult to locate counterparties and execute large orders without order splitting. In addition, fragmentation may give rise to so-called "ghost liquidity," whereby orders are duplicated across venues to increase execution probability but are rapidly canceled on other venues once a trade is executed, reducing the reliability of displayed liquidity. Extra layer of complexity to this problem is added by the fact that more liquid stocks are more likely to get fragmented at the first place.

On July 1, 2019, amid a political dispute over the Swiss–EU institutional framework agreement, the European Commission allowed Switzerland's stock market equivalence to lapse. In response, Switzerland introduced a protective measure prohibiting the trading of Swiss-listed shares on EU venues. As a result, trading in Swiss equities shifted overnight from a fragmented, pan-European environment to a centralized market concentrated on the SIX Swiss Exchange. In February 2021, following the UK's exit from the EU, equivalence between Switzerland and the UK was reinstated. This allowed trading in Swiss shares to resume on UK venues, leading to a re-fragmentation of the market on February 8, 2021. Together, these two events constitute a unique natural experiment: while trading technology and infrastructure remained largely unchanged, the market structure shifted abruptly, enabling a clean identification of the effects of fragmentation on market outcomes.

In this exam, you will investigate how changes to market fragmentation affected stock liquidity.[1]

Design an empirical strategy to answer the research question given the data you have.

Try to address these questions/statements specifically in your report:

- Describe your data and variables used (Table 1)
- Is there a difference between Swiss-listed stocks (SMI index constituents) and stocks listed on French and German exchanges (CAC40 and DAX30 constituents) prior to the event (Table 2)?
- Did the consolidation of liquidity on SMI have an effect on stock liquidity (Table 3)?
- You are worried about unobserved factors/events happening at the same time as the equivalence lapse. Please give some concrete examples of these unobserved factors and discuss whether your empirical design helps mitigate these concerns.
- You are worried that SMI stocks are fundamentally different from the CAC40/DAX30 stocks. Please give concrete examples of potential problems and how this would affect your results if not taken care of. What empirical techniques do you use to address these issues?

[1] E.g., you need to decide on which liquidity measure to use: e.g, quoted bid-ask spread, turnover, Amihud illiquidity (a ratio of absolute return to dollar trading volume), etc.

Data

- isins.csv:
  - ISIN: Stock identifier
  - Index: An identifier of the exchange on which a security is listed
  - Company Common Name
  - Country of Headquarters
  - TRBC Industry Name
- chf_eur.csv:
  - date
  - chf_eur: CHF/EUR exchange rate
- firm_info.csv:
  - date
  - ISIN: Stock identifier
  - ask: Closing ask price
  - bid: Closing bid price
  - book_val_per_share: Book value per share
  - high_price: Maximum price during the day
  - low_price: Minimum price during the day
  - prc: Closing price
  - return_index: Performance of an asset assuming the reinvestment of dividends or other distributions
  - shrout: Shares outstanding (in 1000 shares)
  - tot_assets: Total assets
  - tot_debt_prop: Total debt as proportion of common equity
  - volume: Trading volume (in 1000 shares)

Additional guidance

- Please use information and data as provided in the exam. In other words, do not gather additional data.
- Using code found on the internet is allowed, as long as you reference the sources correctly.
- Plagiarism in any way is not allowed.
- Do not be discouraged if data is fairly messy (i.e., requires additional cleaning), it usually is.
- Try to understand the data before you decide on an econometric approach.
- You need to make informed and motivated choices for variables and models used. Use economic intuition rather than maximizing statistical significance or R^2.
- Make sure that your results are not driven by outliers in your suggested variables.
- Make sure that you decide on variables and data sets you want to use in your model, before starting to code.
- If you have questions during the exam, do not email me, but rather post your question under the respective thread on the Discussion Board on Canvas. This ensures that information is public and distributed fairly. Note that I will exercise my own judgment which questions to answer.
- We encourage students facing problems with R to utilize the Virtual Desktop Infrastructure that NHH offers.
- Good luck!